1. **Scenario:** A system checks if a user is eligible to vote based on their age.
   Write logic to ask the user for their age and determine if they are eligible to vote based on whether they are 18 or older.

   First ask the enter their age.
   Check if age is greater than 18 or older.
   If yes print “Eligible to vote”
   Else print “Not Eligible vote”

2. **Scenario:** A program processes a list of numbers and needs to find the largest value.
   Write logic to identify and return the largest number from a given list.

   Read the list number
   Assume the first number is largest
   comparing each number with the current largest value in the list
   if the largest number is found then
   return the largest number from a given list.

3. **Scenario:** A company provides employees with a 10% bonus if their salary exceeds $50,000.
   Write logic to determine the bonus amount based on the given salary.

   Read the employee’s salary.
   if the salary is greater than $ 50,000
   Calculate of 10% bonus
   otherwise No bonus
   Return the calculated bonus amount

4. **Scenario:** A program evaluates a number to determine if it is even or odd.
   Write logic to check whether a given number is even or odd.
   Read the input number

First check the number is divisible by 2

if yes Print “Even”

Otherwise print “Odd”

5. **Scenario:** A text-processing tool reverses a given word or sentence for formatting purposes.

   Write logic to take a word or sentence as input and produce its reversed version.

   Read the input is word or sentence

   convert the input into list of characters

   reverse the order of character

   Join the reverses character into string

   return the reverses word or sentence

6. **Scenario:** A grading system determines whether a student has passed or failed based on their score.

   Write logic to check if a student has passed a subject by scoring at least 40 marks.

   Read the students mark.
   if the marks are <=40
   Then print “Pass”
   otherwise print “Fail”

7. **Scenario:** A retail store offers a 20% discount if a customer’s total order exceeds $100. Write logic to calculate the final amount to be paid after applying the discount.

   Read the total amount number.
   if the  amount greater than $ 100 calculate 20%
   Subtract the discount from the total amount
   Return the final amount to be paid

8. **Scenario:** A banking system processes **withdrawal** requests and ensures the user has enough **balance**.

Write logic to check if a user has enough balance before allowing a withdrawal and update the remaining balance accordingly.

Read the account balance and withdrawal amount.
If the withdrawal amount is less than or equal to the balance, process the withdrawal
Subtract the withdrawal amount from the balance and return the updated balance
If the withdrawal amount exceeds the balance, print "Insufficient funds"

9. **Scenario:** A calendar system verifies whether a given year is a leap year based on standard leap year rules.

   Write logic to determine whether a given year is a leap year.

   Read the input
   A year is a leap year if it is **divisible by 4**.
   if the year is divisible by 100, then it is **not** a leap year.
   If the year is also divisible by 400, then it **is** a leap year.
   Otherwise, it is not a leap year.

10. **Scenario:** A program filters out only even numbers from a given list.

    Write logic to extract and return only the even numbers from a list.

    Read the list of numbers
    Create an empty list to store even numbers
    iterate through list check
    if divisible by 2 print "Even number" add in the new list
    otherwise print "Not a even number"
    Return the list of numbers